

FYSN33 - PHYSICS: APPLIED COMPUTATIONAL PHYSICS AND MACHINE LEARNING

SPH labs report

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Contents

1	Introduction	1
2	Theory.	1
2.1	Project 1.	1
2.2	Project 2.	3
3	Results.	4
3.1	Project 1.	4
3.2	Project 2.	5
4	Discussion	6
4.1	Project 1.	6
4.2	Project 2.	7
5	Conclusion	8
6	Appendix	8

1 Introduction

Computational fluid dynamics plays a central role in modern physics and astrophysics. One widely used mesh-free numerical method is *Smoothed Particle Hydrodynamics* (SPH), which represents fluids by a set of particles and computes physical quantities through kernel-based interpolation. Due to its Lagrangian nature, SPH is particularly well suited for problems involving large deformations, shocks, and complex geometries.

The purpose of this project is to implement the SPH method and apply it to classical test problems in computational physics. In the first part, the one-dimensional Sod shock tube problem is simulated to assess the ability of SPH to reproduce shock waves, rarefaction waves, and contact discontinuities. In the second part, the method is extended to three dimensions and applied to a planetary collision scenario, including self-gravitational interactions.

2 Theory

2.1 Project 1

First, we will work on the Sod shock tube problem, which is a common method to test the accuracy of computational fluid codes. Only after making sure our code works in this problem will we move on to project 2, which is the collision simulation of planets. In the Sod shock tube problem we have a long straight tube filled with gas, and in the initial state this gas is separated by a membrane in two parts with different physical characteristics. The initial conditions can be found in Table 1., for the left (first row) and right (second row) side of the tube.

Table 1: Initial conditions for Sod's problem in the range $-0.6 \leq x \leq 0.6$.

	ρ	$\mathbf{v}$	e	p	δx	m	No. of Particles
$x \leq 0$	1	0	2.5	1	0.001875	0.001875	320
$x > 0$	0.25	0	1.795	0.1795	0.0075	0.001875	80

In this table ρ is the density, $\mathbf{v}$ is velocity, e is internal energy, p is pressure, δx is particle spacing, and m is the particle mass. After setting up these initial values, we take away the membrane, and in that moment the particles will start to move towards each other. The following things will happen in our tube by taking away the membrane: a shock wave will be produced, which will move into the region of lower density; a rarefaction wave will move into the region of higher density; and a contact discontinuity, which will form in the center and go into the low density region behind the shock. We will be able to see the consequences of these different waves in our plots.

In this project, we will use the SPH (Smoothed Particle Hydrodynamics) method. It is a mesh-free numerical method that simulates fluids by representing them as moving particles. Basically, instead of computing fluid variables at certain positions or grid cells, SPH treats the fluid as a set of particles that carry the physical properties. Here, any field quantity (for example, velocity) is approximated by a weighted average of neighboring particles. This will be done with the use of a smoothing kernel, which will spread each particle's influence

smoothly in space. There are several versions of the smoothing kernel function W_{ij} , but here we will use the cubic spline form. For 1 dimension it is defined as the following:

$$W_{ij} = W(q_{ij}, h) = \frac{1}{h} \begin{cases} \frac{2}{3} - q_{ij}^2 + \frac{1}{2}q_{ij}^3, & 0 \leq q_{ij} < 1, \\ \frac{1}{6}(2 - q_{ij})^3, & 1 \leq q_{ij} < 2, \\ 0, & q_{ij} \geq 2, \end{cases} \quad (1)$$

where:

$$q_{ij} = \frac{|x_i - x_j|}{h}, \quad (2)$$

and h is the smoothing length, and in this project we will approximate it by a constant value, due to the small numbers of particles used. This estimate can be made by the following equation:

$$h = \eta \left(\frac{m_i}{\rho_i} \right)^{\frac{1}{d}} \quad (3)$$

In our motion equations the derivative of the kernel function will also appear, so we need to define that too:

$$\nabla_i W_{ij} = \frac{\partial W_{ij}}{\partial x_i} \quad (4)$$

$$\nabla_i W_{ij} = \frac{dW}{dr} \frac{x_i - x_j}{|x_i - x_j|} \quad \text{with} \quad r = |x_i - x_j| \quad (5)$$

$$\frac{dW}{dr} = \frac{1}{h^2} \begin{cases} -2q_{ij} + \frac{3}{2}q_{ij}^2, & 0 \leq q_{ij} < 1, \\ -\frac{1}{2}(2 - q_{ij})^2, & 1 \leq q_{ij} < 2, \\ 0, & q_{ij} \geq 2. \end{cases} \quad (6)$$

The SPH method has several advantages, and is an optimal choice for a problem like this. It handles large deformations more easily because there is no grid, and is a good alternative for free surfaces and multiphase flows.

All the particles in the tube will move corresponding to these four equations:

$$\rho_i = \sum_{j=1}^N m_j W_{ij} \quad (7)$$

$$\frac{d\mathbf{v}_i}{dt} = - \sum_{j=1}^N m_j \left(\frac{p_i}{\rho_i^2} + \frac{p_j}{\rho_j^2} + \Pi_{ij} \right) \nabla_i W_{ij} \quad (8)$$

$$\frac{de_i}{dt} = \frac{1}{2} \sum_{j=1}^N m_j \left(\frac{p_i}{\rho_i^2} + \frac{p_j}{\rho_j^2} + \Pi_{ij} \right) (\mathbf{v}_i - \mathbf{v}_j) \cdot \nabla_i W_{ij} \quad (9)$$

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{v}_i \quad (10)$$

These are SPH particle equations representing Navier-Stokes dynamics. Equation 7. is the density equation as an SPH approximation, while equation 8. corresponds to the Navier-stokes momentum equation. Equation 9. is the SPH version of the internal energy equation, and equation 10. is just the basic kinematic relation. In equation 8. Π_{ij} means the viscosity, and it is calculated in the following way:

$$\Pi_{ij} = \begin{cases} \frac{-\alpha_{\Pi} \bar{c}_{ij} \phi_{ij} + \beta_{\Pi} \phi_{ij}^2}{\bar{\rho}_{ij}}, & \text{if } \mathbf{v}_{ij} \cdot \mathbf{x}_{ij} < 0, \\ 0, & \text{if } \mathbf{v}_{ij} \cdot \mathbf{x}_{ij} \geq 0 \end{cases} \quad (11)$$

where:

$$\phi_{ij} = \frac{h_{ij} \mathbf{v}_{ij} \cdot \mathbf{x}_{ij}}{|\mathbf{x}_{ij}|^2 + \varphi^2}, \quad \mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j \quad \text{and} \quad \mathbf{x}_{ij} = \mathbf{x}_i - \mathbf{x}_j \quad (12)$$

$$\bar{\rho}_{ij} = \frac{1}{2}(\rho_i + \rho_j) \quad (13)$$

$$\bar{c}_{ij} = \frac{1}{2}(c_i + c_j) \quad (14)$$

$$h_{ij} = \frac{1}{2}(h_i + h_j) \quad (15)$$

which are the average density, speed of sound and smoothing length. Here we will use $\alpha_{\Pi} = 1$, $\beta_{\Pi} = 1$, and $\phi_{ij} = 0.1h_{ij}$ to avoid divergence. In our project first we will compute the viscosity-free version (where $\Pi_{ij}=0$), which corresponds to the Euler case, and then later add viscosity based on equation 11.

The equation state of ideal gas is true for all of the particles in our model:

$$p = (\gamma - 1)\rho e \quad (16)$$

In project 2 we will compute the internal energy based on equation 16. The speed of sound is calculated via the following equation:

$$c = \sqrt{(\gamma - 1)e} \quad (17)$$

where $\gamma = \frac{c_P}{c_V} = 1.4$ is the ratio of specific heat. These are all of the equations that we implemented in project 1.

2.2 Project 2

In project 2, we will do a simulation of colliding two Jupiter-like planets, with the initial conditions being in given datasets. We will use all the formulas already used for Project 1, since it's based on the same SPH model. Nevertheless it is important to point out that there are some differences that need to be implemented for Project 2. The most important is that our equations (and our code) need to be updated from 1 dimension to 3 dimensions, since we are working with planetary collisions. This means that position and velocity become vectors, and the smoothing kernel function W_{ij} will be multiplied by a different constant instead of

$1/h$ (which corresponded only to the one dimensional case). The updated smoothing kernel function is the following:

$$W_{ij} = W(q_{ij}, h) = \frac{3}{2\pi h^3} \begin{cases} \frac{2}{3} - q_{ij}^2 + \frac{1}{2}q_{ij}^3, & 0 \leq q_{ij} < 1, \\ \frac{1}{6}(2 - q_{ij})^3, & 1 \leq q_{ij} < 2, \\ 0, & q_{ij} \geq 2, \end{cases} \quad (18)$$

$$\frac{dW}{dr} = \frac{1}{h} \cdot \frac{3}{2\pi h^3} \begin{cases} -2q_{ij} + \frac{3}{2}q_{ij}^2, & 0 \leq q_{ij} < 1, \\ -\frac{1}{2}(2 - q_{ij})^2, & 1 \leq q_{ij} < 2, \\ 0, & q_{ij} \geq 2. \end{cases} \quad (19)$$

The other big update that we need to implement in our code is the addition of gravity. We will need to add a gravitational term to equation 8. This term is the following:

$$\left(\frac{d\mathbf{v}}{dt}\right)_{\text{Gravity}}^i = -\frac{G}{2} \sum_j^N m_j (\nabla_i \phi_{ij}(h_i) + \nabla_i \phi_{ij}(h_j)) \frac{d\mathbf{x}}{r_{ij}} \quad (20)$$

where

$$\frac{d\phi(r, h)}{dr} = \begin{cases} \frac{1}{h^2} \left(\frac{4}{3}R - \frac{6}{5}R^3 + \frac{1}{2}R^4 \right) & 0 \leq R < 1, \\ \frac{1}{h^2} \left(\frac{8}{3}R - 3R^2 + \frac{6}{5}R^3 - \frac{1}{6}R^4 + \frac{1}{15R^2} \right) & 1 \leq R < 2, \\ \frac{1}{r^2} & R \geq 2 \end{cases} \quad (21)$$

and

$$R_{ij} = \frac{r_{ij}}{h} = \frac{|\mathbf{x}_i - \mathbf{x}_j|}{h} = \frac{|d\mathbf{x}|}{h} \quad (22)$$

Once we have a stable planet defined by these equations, we will copy the initial variables of that one and add a shift in the position, and this way we get the second planet. Then we can make these two planets move towards each other, and simulate a collision. After that's working, we can add spin to the planets, which will be an additional term to the initial velocity:

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} \quad (23)$$

where $\boldsymbol{\omega}$ is the angular velocity.

3 Results

3.1 Project 1

For part 1, we begin by defining all initial values for both sides from Table (1) and assembling them into arrays. We then define the constants: the small fudge factor $\eta = 1.5$, $\alpha_{\Pi} = 1$, and $\beta_{\Pi} = 1$. We use a constant smoothing length h (15), which is calculated by taking the left- and right-side values and computing their average.

Next, we define the necessary functions. We start with the kernel and its derivative (1 and 6), followed by the expressions for density, pressure, and sound speed (7, 16, and 17), which are explicitly given in the lab manual.

Finally, we implement a function that computes both the time derivative of the internal energy and the acceleration (i.e., the velocity derivative). The artificial viscosity is calculated inside this function using the constants α_{Π} and β_{Π} , together with the average density (7) and sound speed (17). The term ϕ_{ij} is computed using the constant smoothing length h and $\varphi = 0.1h$, along with the relative position vector x_{ij} and velocity vector v_{ij} . These vectors are obtained by broadcasting the 1D arrays. The remaining quantities are then computed directly following Eqs. (8) and (9), using the particle masses, the kernel derivative, density, and pressure.

On solving the above equations, we get the following evolutions of the quantities:

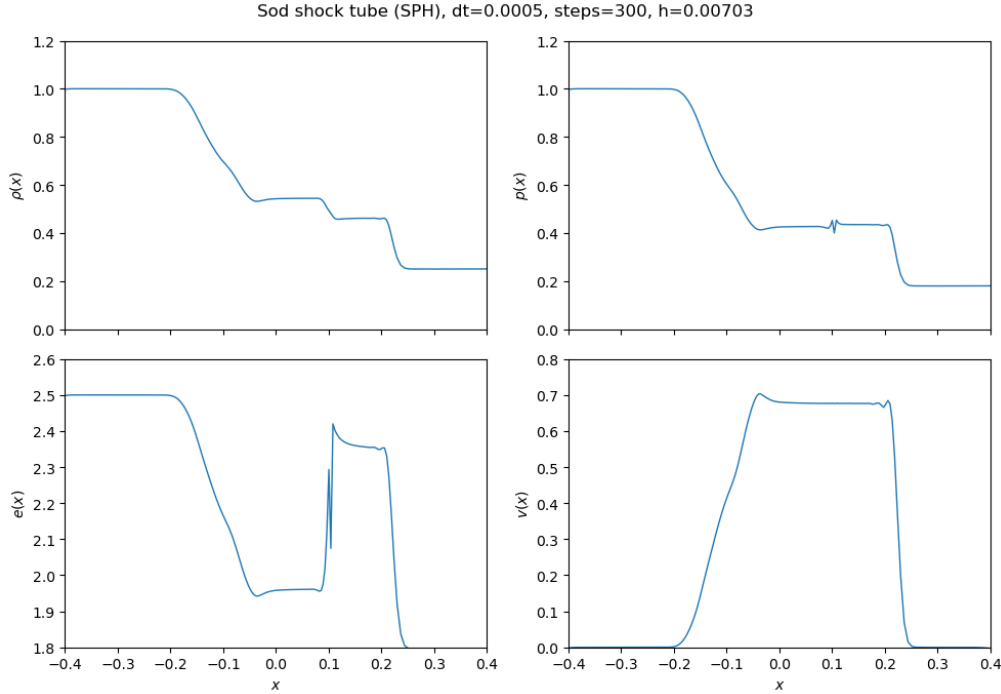


Figure 1

Results for Sod shock tube simulation, (clockwise from top left - density, pressure, velocity and energy)

3.2 Project 2

This is similar to project 1, except now there are two more dimensions and an additional gravitational term to compute (20). For this part, we use the file `Planet600.dat`, which contains 600 particles per planet. First, we made sure the code was working for one planet. After that, we duplicated the data so that the second planet has the same initial conditions, and then made a shift in its position. This way the two planets could go flying towards each other, and collide in the center. Adding spin to the planets was the next step in the project. We added the spin term based on Equation 23. to the initial velocity. The angular velocity ω is calculated using Jupiter's orbital period. The initial velocity is set equal to the

escape velocity and is directed along the x -axis. The initial positions of the planetary centers are defined as $2.8R$ and $0.5R$ in the x and y directions, respectively, where R denotes the planetary radius. For each time step, the system is advanced by a time increment $dt = 2$, and the simulation is run for a total of 2400 steps.

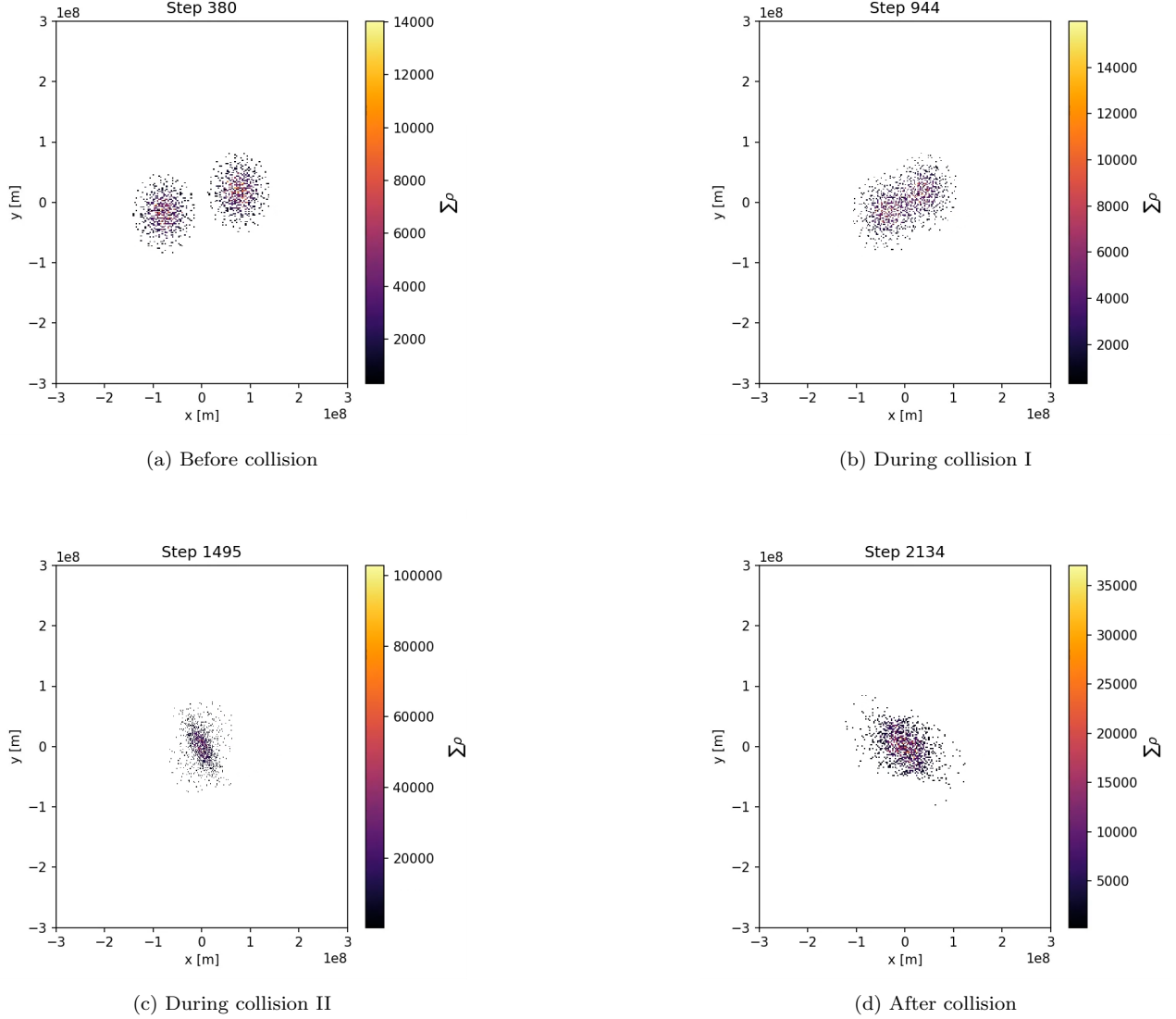


Figure 2: Simulation of two Jupiter-like planets

4 Discussion

4.1 Project 1

Initially, the fluids started out as though they were separated by an 'invisible non permeable barrier'. The fluid on the left has a higher density, higher pressure and higher energy particles than the one on the right.

Once the barrier is removed, the fluids start to intermix, higher pressure fluid on the left tries to flow towards the right. On the density plot (Figure 1.), this results in a smooth slope wherein density keeps decreasing towards the right. The pressure also decreases smoothly in this region. The velocity of the particles increases in this region due to their movement. The peak in density start falling as time progresses until it collapses completely (when the fluid eventually moves out of the tube).

The second region (between -0.1 to 0.1) is the area where fluids from both sides meet, and it is marked by a nearly stable pressure, energy and velocity. The density plot shows a stepped region at the 'boundary'. This region moves to the right with time, with the region being almost unchanged.

The third region has a sharp drop in density, where the higher pressure fluid tries to compress the lower pressure fluid. Thus, there is a sharp difference in pressure and energy. This is the shock region, which keeps moving to the right with time.

The absence of boundary conditions at both ends causes the fluid to eventually drain out of the tube, i.e. everything settles down to zero in a few seconds.

4.2 Project 2

Our results in project 2 can be seen on Figure 2. When the two planets are first generated, the forces due to pressure and gravity act on the particles. This leads to a situation where the particles near the core 'push' themselves outward due to pressure being the stronger force and the particles at the crust, being more affected by gravitational force are 'pulled' inwards towards the core. This leads to the planet 'breathing' due to the balancing of the two forces. In the simulation, one of the planets is made to collide against the other by giving it a certain velocity. The velocity determines whether after the collision, the planets will remain or be disintegrated into a debris field.

After the collision happened with our implemented velocity the planets eventually disintegrated into a debris field going outward with time. For lower collision velocities however, the planets 'fused' together to form a superplanet, with the debris field being much smaller and most of it being pulled into the newly formed planet.

There can be a third case too, wherein after the collision both of the planets remain, with one being comparatively bigger than the other. If the parameters are right, the simulation will show a natural satellite formation, similar to the 'Giant Impact Hypothesis'.

Some approximations were needed to be made to simplify our code. For example, for the smoothing length h we choose a constant value, and not a variable based on Equation 3. For real simulations, the value of h should be calculated according to that. Another parameter that could be considered an approximation is the number of particles used per planet. A few hundred particles is orders of magnitude smaller than what is used in realistic models. The equation state of ideal gas (Equation 16.) is also a simplification, since real planets are not only made up of gas, so more specific equations should be used in real life examples. The SPH method in itself is also an approximation, although this method is often used in many planetary collision simulations, making it a good model for these type of problems.

5 Conclusion

In project 1, we successfully implemented the Sod shock tube problem based on the Smoothed Particle Hydrodynamics (SPH) model with our code. In this model we have a tube filled with gas, in which a membrane separates the two sides from each other: the left side with higher, the right side with lower density. We take away this membrane from the tube, and the particles will start moving towards each other, described by the equations mentioned in the theory part. On Figure 1. we can see our results, which are as expected. For example on the density plot we can clearly see the shock wave (right side), the rarefaction wave (left side), and contact discontinuity (in the middle), which were earlier mentioned in the theory part.

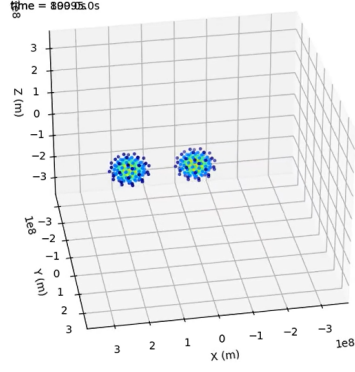
In project 2, we simulated the collision of two planets. We used the same type of equations as in project 1, based on the same SPH model (with the addition of a gravity based acceleration). Our results can be seen on Figure 2 . In these plots, the different stages of the collision (approach, interaction, disruption) are realistic and meet our expectations. The density evolution and particle redistribution are also physically reasonable.

These results demonstrate that SPH is capable of simulating large-scale hydrodynamical interactions. The approximations we used did not spoil the general picture of the results, but in real-life planetary collision simulations some of the parameters (for example, the smoothing length, larger number of particles) should be more accurately defined.

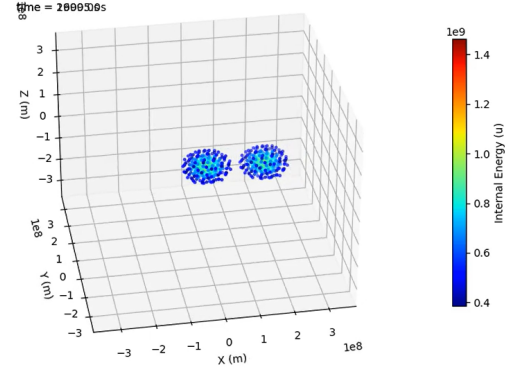
6 Appendix

The simulation was also visualized with a 3d plot. The simulation was an animation which could not be embedded into this document, but there are some images of the collision at different time steps. ¹

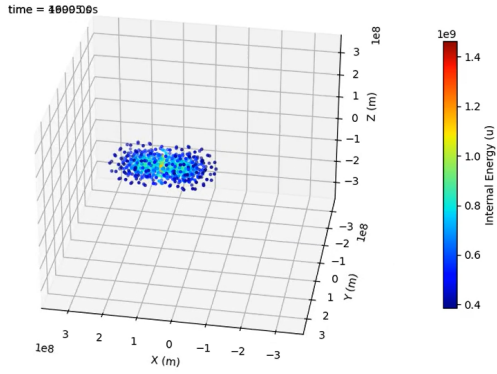
¹These plots (3) were produced from a different code than the uploaded one.



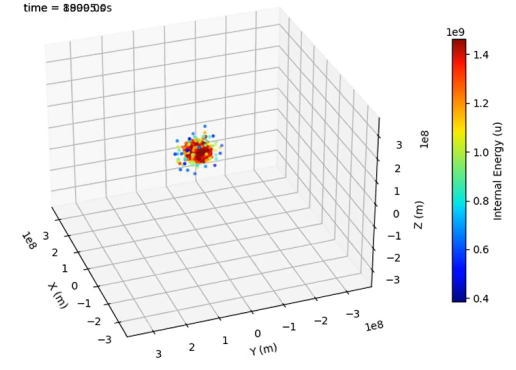
(a) Initialization



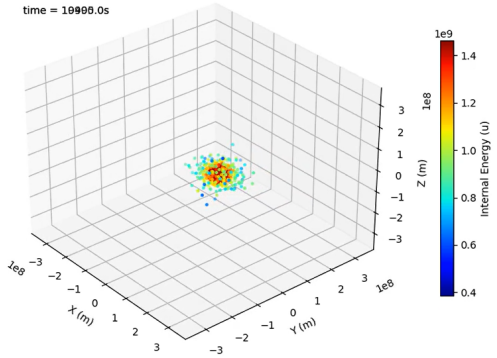
(b) Start of collision process



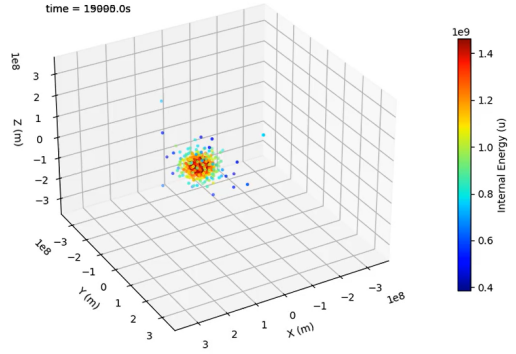
(c) Collision



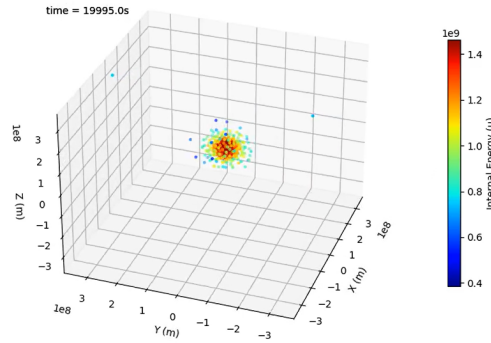
(d) Fused into one superplanet



(e) Initial debris field



(f) Debris field



(g) Debris field settles down

Figure 3: Simulation of collision of two planets